PSFD-Net: Progressive Supervision-Based Frequency-Domain Decoupling Network for Hyperspectral and Multispectral Image Fusion

Yan Mou, Genji Yuan, Jintao Song, and Jinjiang Li

***Abstract*—Hyperspectral–multispectral image fusion (HMIF) aims to reconstruct high-fidelity, high spatial-resolution hyperspectral images (HR-HSIs) by combining the complementary features of both modalities. However, the low spatial resolution of low-resolution hyperspectral images (LR-HSIs) and the sparse spectral sampling of high-resolution multispectral images (HR-MSIs) limit the representation of complementary features. Spatial-domain methods still suffer from spatial and spectral distortions, which stem from the coupling between spatial and spectral features in the spatial domain, making it difficult to decouple and calibrate complementary features and ultimately affecting fusion accuracy. To address this issue, we tackle the feature coupling problem from a frequency-domain perspective. Specifically, we propose a progressive supervision-based frequency-domain decoupling network (PSFD-Net). The core idea of PSFD-Net is to decouple, calibrate, and effectively fuse high- and low-frequency components in the frequency domain, while imposing training constraints to facilitate high-fidelity fusion reconstruction. We first introduce an adaptive frequency-decoupling module (AFDM) to decompose features into high- and low-frequency components, and then design a physically aware feature calibration (PAFC) module to calibrate and enhance frequency components under physically aware constraints. Finally, a full-frequency guided spatial–spectral fusion (FGSS) Module is employed to effectively integrate frequency information under full-frequency guidance. In addition, we design a progressive supervision strategy based on an annealing mechanism to achieve high-fidelity reconstruction from global spectral information to local structural details. Extensive experiments demonstrate that the proposed PSFD-Net outperforms current state-of-the-art methods.**

***Index Terms*—Deep learning, Fourier transform, frequency-domain decoupling, hyperspectral and multispectral image fusion, progressive supervision.**

I. Introduction

HIGH spatial-resolution hyperspectral images (HR-HSI) simultaneously capture fine spatial structures and hundreds of contiguous spectral bands, enabling precise characterization of land cover in both spatial and spectral dimensions. Consequently, HR-HSI has important applications in scene classification [1], object detection [2], [3], anomaly detection [4], and semantic segmentation [5]. However, due to the inherent tradeoff between signal-to-noise ratio and spatial–spectral resolution in sensors, directly acquiring HR-HSI is challenging. To overcome this limitation, hyperspectral–multispectral image fusion (HMIF) has been proposed, aiming to combine the rich spectral information of low-resolution hyperspectral images (LR-HSIs) with the fine spatial details of high-resolution multispectral images (HR-MSIs) to reconstruct high-fidelity HR-HSI [6]. LR-HSI provides complete spectral information but low spatial resolution, whereas HR-MSI offers high spatial resolution but sparse spectral coverage. Extracting reliable complementary features from these sources while achieving high-fidelity reconstruction remains a core challenge in the field.

Early HMIF methods relied on handcrafted priors, including component substitution (CS) and multiresolution analysis (MRA) [7], [8], [9]. CS assumes linear separability between spatial and spectral components and reconstructs HR-HSI by replacing intensity components, while MRA extracts high-frequency details through techniques such as wavelet transforms for global fusion. These approaches are limited by their linear assumptions and cannot model the complex nonlinear spectral mixing present in real scenes, thus restricting fusion accuracy [10]. With the rise of deep learning, convolutional neural networks (CNNs) and transformers have become the dominant architectures for HMIF [11], [12], [13], [14], [15], [16], [17], [18]. CNNs capture local spatial details through receptive fields, whereas Transformers leverage global self-attention to model long-range spectral dependencies. Unlike traditional methods, these data-driven models can learn end-to-end nonlinear mappings and implicitly explore spatial–spectral relationships in the feature space. Despite significant advances, the coupling between spatial and spectral features persists, making it difficult for models to fully separate their complementary information, which restricts the high-fidelity reconstruction of images.

Specifically, as illustrated in Fig. 1(a), the spectral values of LR-HSI exhibit deviations due to its limited spatial resolution, particularly in edge regions due to spatial mixing effects. Meanwhile, due to its limited spectral resolution, HR-MSI suffers from sparse spectral sampling, which results in the loss of fine spectral details and consequently intro-

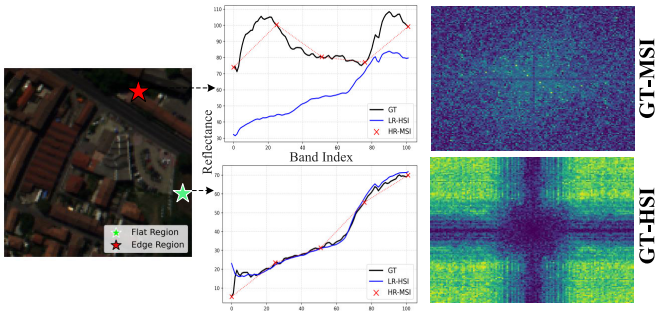
Received 14 February 2026; revised 7 April 2026; accepted 27 April 2026. Date of publication 8 May 2026; date of current version 19 May 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 61772319, Grant 62002200, and Grant 62576193; in part by the Natural Science Foundation of Shandong Province under Grant ZR20250C695, Grant ZR20250C712, and Grant ZR20250C985; and in part by the Scientific Research Start-Up Fund Project of Shandong Technology and Business University under Grant BS202424. *(Corresponding author: Genji Yuan.)*

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Digital Object Identifier 10.1109/TGRS.2026.3691537

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(a)Spatial-Spectral Analysis (b)Amplitude Analysis

Fig. 1. Modality limitations and frequency-domain analysis on the Pavia dataset. (a) Spectral profile comparison of sample points reveals that while LR-HSI maintains spectral continuity, it exhibits noticeable radiometric intensity biases, particularly in edge regions. Conversely, HR-MSI provides accurate intensity but suffers from sparse sampling; the missing sampling points (indicated by dashed lines) fail to characterize the complete spectral variations (with larger deviations at edges), thereby inducing nonphysical structural artifacts. (b) Amplitude error maps demonstrate that errors in HR-MSI are primarily concentrated in the low-frequency components, whereas errors in LR-HSI are dominant in the high-frequency regions. This indicates that the low-frequency information of LR-HSI and the high-frequency information of HR-MSI possess higher physical reliability.

duces detail deviations. This effect is particularly evident in edge regions, where spectral mixing effects lead to noticeable artifact interference. Traditional spatial-domain methods struggle to independently disentangle and calibrate this limitation, which stems from the intrinsic coupling of features within the spatial domain. In the spatial domain, the spectral properties of materials are tightly coupled with spatial gradients under the pixel coordinate system [19]. Within spatial-domain feature processing frameworks, such as CNNs, spectral processing inevitably affects spatial gradient sharpness, while enhancing spatial details can degrade spectral fidelity, ultimately compromising fusion quality. Moreover, neural networks tend to fit low-frequency components preferentially, resulting in the loss of high-frequency details [20]. Standard losses like mean squared error (mse) are insensitive to high-frequency information, making it difficult to recover fine spatial details and limiting reconstruction accuracy. Therefore, effectively extracting, calibrating, and reconstructing complementary features to mitigate the negative impacts of modality limitations remains a critical challenge in HMIF.

To address this issue, we analyze the extraction and reconstruction of complementary features from a frequency-domain perspective. Low-frequency components concentrate image energy and represent the global spectral distribution of materials, while high-frequency components carry gradient variations and encode spatial details [21]. As shown in Fig. 1(b), the low-frequency components of LR-HSI and the high-frequency components of HR-MSI exhibit high similarity with the ground truth (GT), indicating that global spectral information is primarily supported by LR-HSI's low-frequency components, whereas fine spatial details are mainly contributed by HR-MSI's high-frequency components. However, LR-HSI's low-frequency components suffer from intensity deviations, and HR-MSI's high-frequency components are contaminated with significant structural artifacts. Given the complementary advantages of the two modalities in spatial and spectral resolution, these components can be calibrated and enhanced using cross-modality priors to achieve high-fidelity fusion. The annealing mechanism dynamically adjusts the annealing factor, enabling the model to progressively strengthen constraints on different features during training [22], [23]. Accordingly, it is incorporated into the loss function, allowing the model to first stabilize low-frequency spectral components and then progressively reconstruct high-frequency details. This compensates for high-frequency information neglected by mse, enabling the collaborative optimization of spectral and spatial features and improving reconstruction fidelity.

Based on the above analysis, we propose the progressive supervision-based frequency-domain decoupling network (PSFD-Net) to address the feature coupling problem in the spatial domain and achieve high-fidelity HR-HSI reconstruction. The core idea of PSFD-Net is to decouple high- and low-frequency features and further integrate them using physically aware calibration and collaborative fusion strategies to effectively combine complementary information. The network consists of three key modules: adaptive frequency-decoupling module (AFDM) for separating high- and low-frequency components, physically aware feature calibration (PAFC) for calibrating and enhancing these components, and full-frequency guided spatial–spectral fusion (FGSS) for their effective integration. In addition, a progressive supervision strategy ($\mathcal{L}_{PS}$) is introduced, where an annealing factor $\lambda(t)$ guides the model from spectral alignment to structural sharpening in a smooth manner, ensuring that the fused images maintain both high spectral fidelity and fine spatial details.

Our contributions are as follows.

1) We reveal the impact of inherent modality limitations on high-fidelity fusion and point out that spatial-domain methods struggle to disentangle highly coupled spectral–spatial features, which prevents them from effectively overcoming these negative impacts and thus restricts reconstruction accuracy and provides a new problem perspective for the HMIF task.
2) We propose PSFD-Net, which disentangles spatial and spectral features in the frequency domain. AFDM and PAFC are used to decouple and calibrate-enhance high- and low-frequency features, respectively; FGSS is designed to effectively integrate features for high-fidelity spatial–spectral reconstruction.
3) We propose an annealing-based loss, $\mathcal{L}_{PS}$, which guides the model to first optimize low-frequency spectral features and then refine high-frequency spatial details, balancing spatial structure refinement and spectral fidelity. Experimental results demonstrate that our method achieves state-of-the-art performance on four public datasets.

II. Related Work

A. Traditional Methods

Traditional HMIF methods primarily rely on handcrafted priors and mathematical optimization, and can be broadly categorized into two types [24], [25], [26], [27], [28]. The

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first type is based on transform-domain and feature injection methods, including MRA, CS, and variational statistical models, which inject details by mapping images to auxiliary spaces, thereby enhancing spatial resolution while preserving the original spectral characteristics. For example, Akhtar et al. [29] introduced sparse representation into a Bayesian framework, guiding sparse coding with an overcomplete dictionary learned from LR-HSI, significantly improving reconstruction robustness under complex noise. The second type is based on physical degradation modeling, mainly involving matrix/tensor decomposition, which models the degradation process using the physical principles of remote sensing imaging. Yokoya et al. [24] employed a linear spectral mixing model to decompose features into endmembers and abundances, and achieved high-fidelity cross-modal reconstruction via alternating updates. Although these traditional methods offer good physical interpretability, their heavy reliance on handcrafted priors and linear assumptions limits their ability to capture nonlinear mappings in real scenarios, often resulting in spectral distortions or blurred details [30], [31].

B. Deep Learning Methods

In recent years, with the rapid development of deep learning, CNNs and transformers have become the dominant approaches in the HMIF field due to their strong nonlinear mapping capabilities [32], [33], [34], [35], [36], [37], [38], [39]. Early studies primarily relied on CNNs to extract high-fidelity local spatial–spectral features. For instance, Yuan et al. [40] employed multiscale convolutional kernels to extract features, effectively integrating multiscale complementary information to improve reconstruction accuracy, while Liu et al. [41] designed a dual-branch network to separately extract complementary features, enabling collaborative spatial–spectral fusion, Yang et al. [42] proposed a deep-prior-regularized variational fusion model that jointly leverages the deep prior and the multiscale nonlocal prior of LR-HSI, effectively enhancing both spatial and spectral fidelity. Subsequently, the introduction of Transformers overcame the limited receptive field of CNNs, allowing for global modeling of spatial–spectral dependencies [43]. For example, Wang et al. [44] developed a multiscale cross-attention fusion mechanism to achieve efficient interaction of spatial–spectral information, Ma et al. [45] employed a directional pairwise multihead cross-attention mechanism to facilitate bidirectional complementary information exchange, effectively alleviating the common issues of detail loss and spectral distortion in image fusion. More recently, Chen et al. [46] adopted inverse spectral cross-attention and adaptive interaction strategies to enable deep fusion of complementary features. For CNN-based models, convolutional operators are limited by a fixed receptive field and extract spatial information through the weighted aggregation of neighboring pixels. While this process enhances structural features, it often introduces spectral mixing effects. Similarly, cross-channel convolutions used for spectral fusion can alter pixel distributions, thereby affecting spatial gradients. For transformer-based models, spatial attention models long-range dependencies via global feature aggregation, but cross-location interactions may introduce responses inconsistent with the original spectra. Meanwhile, spectral attention, through cross-channel weighting, can weaken the expression of local spatial structures. Therefore, whether using convolutions or attention mechanisms, the core operations fundamentally rely on feature aggregation and reconstruction, jointly modeling spatial and spectral information within a unified feature space. The lack of explicit decoupling mechanisms makes it difficult to effectively avoid spatial–spectral feature entanglement.

C. Frequency-Domain Decoupling Methods

Recently, joint spatial-frequency modeling has emerged as a highly promising research direction in the field of HMIF. These methods leverage operators such as the discrete Fourier transform (DFT) to establish global perceptual capabilities in the frequency domain, which are integrated with local spatial representations to achieve deep synergy between spatial texture enhancement and spectral energy preservation [47], [48], [49]. For instance, Cao et al. [50] employ a cross-domain perceptual attention mechanism to effectively capture the intrinsic spatial–spectral dependencies. Meanwhile, it integrates a Fourier-domain perception module that leverages amplitude and phase information to enhance cross-stage feature aggregation, thereby significantly improving the accuracy of spatial–spectral reconstruction, [49] significantly strengthens the model's robustness in reconstructing fine high-frequency textures and modeling global contextual dependencies by separating spectral attributes and introducing Gabor convolution operators. Although these methods introduce frequency-domain information to enhance feature representation, they essentially perform auxiliary fusion within a spatial-domain framework, without explicitly distinguishing different frequency components from a frequency decomposition perspective. As a result, spatial and spectral information remain coupled within a unified feature space, making intermodal interference unavoidable and limiting high-fidelity fusion performance. To this end, we address the coupling issue from a frequency-decoupling perspective.

Frequency-domain decoupling decomposes complex signals into independent frequency components with clear physical meaning, providing a foundation for targeted processing of different frequency bands and enhancing feature representation. This strategy has been widely applied in various visual tasks [20], [51]. For instance, Shan et al. [21] separate high- and low-frequency components via the Fourier transform to capture edge details and main structures, improving classification accuracy. Zhao et al. [52] employ frequency-domain decoupling with parallel perception to overcome the detail loss limitation of Transformers, enhancing fine-grained information in remote sensing images. Wang et al. [53] obtain low-, high-, and combined mid-frequency features for adaptive super-resolution enhancement, while Huang et al. [54] extract target-specific and consistent spectral features, suppressing background noise and enabling precise object localization in complex scenes. These studies demonstrate that frequency-domain decoupling can effectively reduce interference between features and provide a physical basis for directed processing

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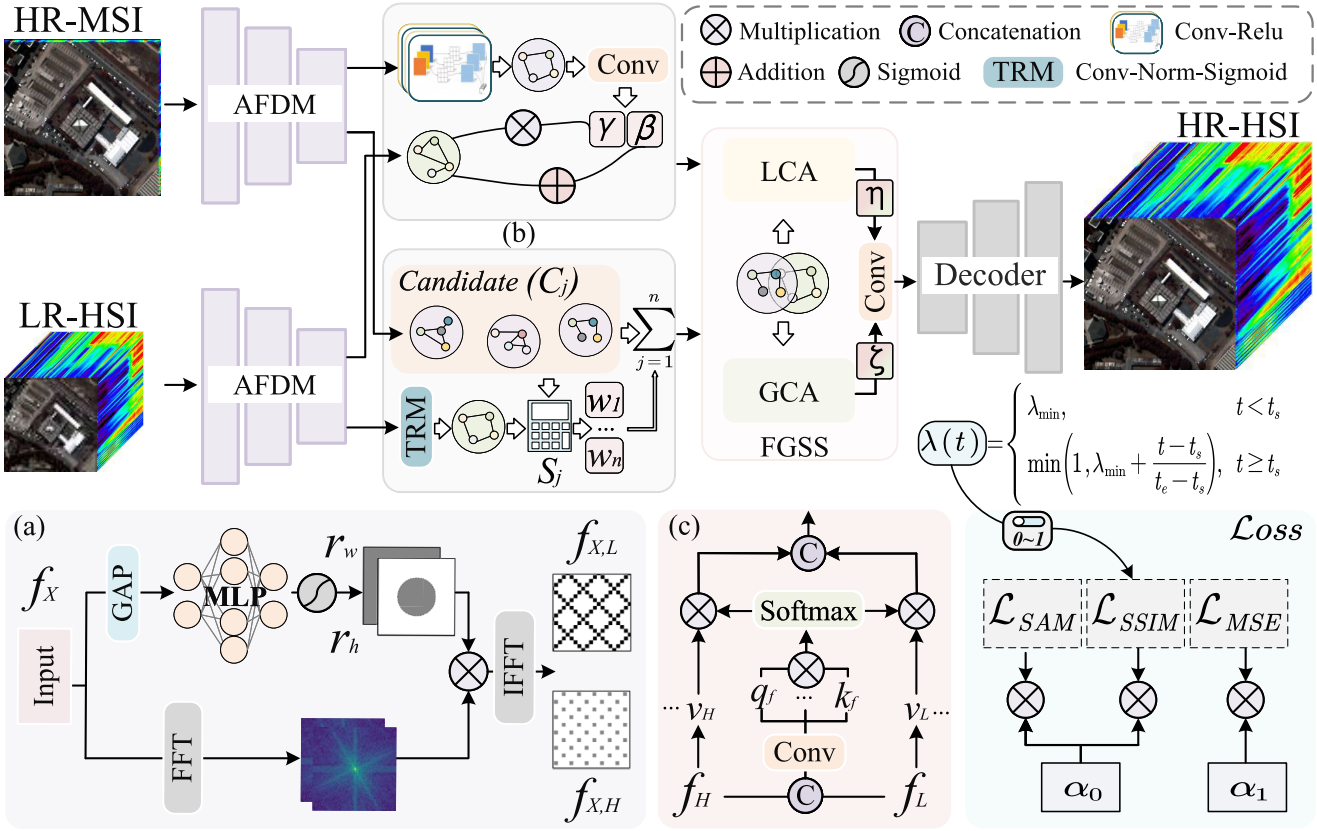


Fig. 2. Overview of the proposed PSFD-Net. The architecture consists of three core stages: 1) AFDM for multiscale frequency decoupling; 2) PAFC for targeted calibration of decoupled features; and 3) FGSS for leveraging full-frequency information to guide the interaction and fusion of complementary features, followed by a decoder to reconstruct the HR-HSI. (a) and (b) Detail the internal structures of AFDM and PAFC, respectively, while (c) illustrates the cross-modal attention mechanism within the FGSS framework. Furthermore, $\lambda(t)$ is employed to modulate the training process, thereby balancing the tradeoff between spatial details and spectral consistency. Abbreviations: AFDM: adaptive frequency-decoupling module; PAFC: physically aware feature calibration module; FGSS: full-frequency guided spatial–spectral fusion module; TRM: texture reference module; LCA: local cross-attention; and GCA: global cross-attention.

of semantically distinct components, significantly improving the flexibility and accuracy of feature representation. However, existing methods often assume that the decoupled components are inherently physically reliable, neglect in-depth analysis of their limitations, and lack targeted calibration mechanisms, which prevents the model from effectively addressing representation biases and thus restricts overall performance.

III. Methodology

A. Overall Architecture

Our goal is to fully exploit and integrate spatial–spectral information for high-quality HR-HSI reconstruction. As shown in Fig. 2, PSFD-Net consists of three stages. First, multiscale features of LR-HSI and HR-MSI are extracted in parallel using two cascaded 3×3 convolutions, denoted as $f_X^i = f_{LR-HSI}^i, f_{HR-MSI}^i$ $(i = 1, 2, 3)$. Next, AFDM dynamically generates frequency filtering masks via a content-aware mechanism to separate features into high-frequency and low-frequency components, while PAFC calibrates them under dual physically aware constraints to capture reliable spatial and spectral information. Finally, FGSS integrates spatial–spectral information through full-frequency guided parallel cross-attention, generating high-fidelity fused features that preserve both global spectral characteristics and fine edge details. The fused features are hierarchically concatenated via residual branches and progressively upsampled by the decoder to reconstruct the high-fidelity HR-HSI. During training, in addition to $\mathcal{L}_{mse}$ for overall reconstruction accuracy, $\mathcal{L}_{PS}$ is introduced. Guided by the dynamic weight $\lambda(t)$, it directs the model to follow a "stabilize spectral first, refine structure later" strategy, achieving an optimal balance between spectral fidelity and spatial detail enhancement.

B. Adaptive Frequency-Decoupling Module

AFDM aims to separate low-frequency and high-frequency information. The spectral responses of ground objects and spatial structures exhibit deep physical coupling in the pixel space [19]. Existing spatial-domain methods struggle to handle this coupling, often introducing spectral distortion when enhancing spatial details, and vice versa, thereby limiting fusion accuracy. To address this issue, we introduce the AFDM, as illustrated in Fig. 2(a). AFDM adaptively decomposes the features of each spectral channel into low-frequency and high-frequency components using the Fourier transform. The high-frequency components represent local spatial structure details, while the low-frequency components carry the core radiometric energy

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distribution across bands, serving as a spectral baseline [21]. The core idea of this design lies in leveraging the orthogonality of frequency subspaces to separate spectral responses and geometric textures that are originally entangled in the spatial domain. On this basis, the bandwidth of the low-pass filter is dynamically calibrated to ensure consistent spectral constraints across all channels. This "divide-and-conquer" strategy provides a foundation for subsequent modules to perform spatial–spectral collaborative modeling on the decoupled features, effectively mitigating the mutual interference between spatial enhancement and spectral fidelity that is common in spatial-domain methods, thereby improving the overall fusion accuracy.

Specifically, we first employ the DFT to transform the stage-i features $f^i_{LR\text{-}HSI}$ and $f^i_{HR\text{-}MSI}$ into the frequency domain, formulated as

$$F\left(f_x^i\right)(u,v) = \frac{1}{\sqrt{H_i W_i}} \sum_{h=0}^{H_i-1} \sum_{w=0}^{W_i-1} f_x^i(h,w)\, e^{-j2\pi\left(\frac{h}{H_i}u + \frac{w}{W_i}v\right)} \quad (1)$$

where u and v denote frequency components. Subsequently, a content-aware adaptive ratio prediction mechanism is introduced to dynamically predict the normalized cutoff ratios (r_h^i, r_w^i)

$$\left(r_h^i, r_w^i\right) = \sigma\left(\mathrm{MLP}\left(\mathrm{GAP}\left(f_x^i\right)\right)\right) \quad (2)$$

where GAP$(\cdot)$ denotes the global average pooling operation and $\sigma(\cdot)$ represents the Sigmoid activation function. This mechanism is based on the intrinsic relationship between global spatial structure and frequency-domain energy distribution: the energy in smooth regions is mainly concentrated in a narrow low-frequency band, while complex texture regions exhibit a wider spectrum. The generated ratio is used to determine the dynamic cutoff boundary of the low-pass filter (R_H^i, R_W^i)

$$R_H^i = r_h^i \cdot \frac{H_i}{2}, \quad R_W^i = r_w^i \cdot \frac{W_i}{2} \quad (3)$$

where (R_H^i, R_W^i) define the boundaries of the low-frequency energy support region in the frequency domain. The dynamic cutoff boundaries allow the determination of the global low-frequency spectral components of the input features, based on which the low-pass filter is constructed; the high-frequency components are obtained as the complement of the low-pass mask, thereby effectively capturing high-frequency spatial details. By leveraging these adaptive boundaries, the filter can separate low- and high-frequency components according to the frequency distribution of the input features, enabling it to HMIF scenarios. Based on this, a learnable low-pass filter $M^i(u,v)$ is constructed as

$$M^i(u,v) = \sigma\left(R_H^i - |u|\right) \cdot \sigma\left(R_W^i - |v|\right). \quad (4)$$

Using this mask, the frequency spectrum is decomposed into mutually independent low-frequency and high-frequency components

$$F^i_{x,\mathrm{low}} = F_x^i \odot M^i(u,v), \quad F^i_{x,\mathrm{high}} = F_x^i \odot \left(1 - M^i(u,v)\right) \quad (5)$$

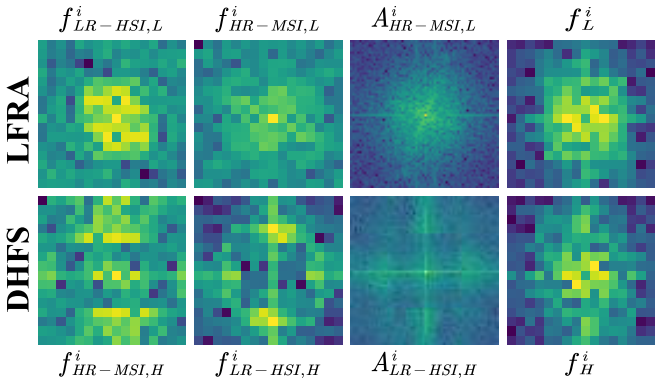


Fig. 3. Amplitude characteristic analysis of the PAFC module on the Houston dataset. The visualization illustrates the calibration process of two key submodules: $f^i_{LR-HSI,L}$ exhibits significant energy diffusion and centroid shift. By anchoring the condensed energy center via the radiometric prior $A^i_{HR-MSI,L}$, the calibrated f^i_L achieves a more concentrated energy distribution. $f^i_{HR-MSI,H}$ is heavily contaminated by nonphysical noise. Through discriminative reorganization guided by the structural prior $A^i_{LR-HSI,H}$, the calibrated f^i_H effectively suppresses noise artifacts while preserving sharp edge topology. Abbreviations: LFRA: low-frequency radiometric alignment; DHFS: discriminative high-frequency feature selection.

where $\odot$ represents element-wise multiplication. Finally, the low-frequency and high-frequency components are mapped back to the spatial domain via the inverse DFT (IDFT)

$$f^i_{x,L} = \mathcal{F}^{-1}\left(F^i_{x,\mathrm{low}}\right), \quad f^i_{x,H} = \mathcal{F}^{-1}\left(F^i_{x,\mathrm{high}}\right). \quad (6)$$

C. PAFC Module

Although AFDM effectively extracts high- and low-frequency components, both still suffer from significant artifacts. As illustrated in Fig. 3, the low-frequency spectrum of $f^i_{LR-HSI,L}$ exhibits obvious energy dispersion and centroid shifting, while the high-frequency part of $f^i_{HR-MSI,H}$ is corrupted by texture noise. These phenomena reveal that the decoupled components remain heavily constrained by the inherent physical limitations of the sensors: the restricted spatial resolution of LR-HSI induces spectral energy bias, whereas the limited spectral resolution of HR-MSI renders its high-frequency structures susceptible to modal interference. Conversely, $f^i_{HR-MSI,L}$ possesses a more concentrated energy center, and $f^i_{LR-HSI,H}$ maintains reliable structural topology. Such cross-modal complementary advantages provide a robust baseline for adaptive calibration, imposing effective constraints to reconstruct high-fidelity spatio-spectral features.

Based on the aforementioned analysis, we introduce the PAFC module [see Fig. 2(b)], which implements a dual physically aware constraint to reconstruct high-fidelity spatio-spectral features: 1) it leverages the spatial distribution priors of HR-MSI to calibrate the energy dispersion in LR-HSI low-frequency components and 2) it leverages the structural benchmarks of LR-HSI to achieve noise suppression and the discriminative recomposition of high-frequency details within HR-MSI.

1) Low-Frequency Radiometric Alignment Module: LFRA is designed to adaptively calibrate the low-frequency energy dispersion of LR-HSI by leveraging the spatial distribution

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priors of HR-MSI. Specifically, it takes the low-frequency components $\{f^i_{LR-HSI,L}, f^i_{HR-MSI,L}\}$ generated by AFDM at each scale i as inputs. First, the spatial prior distribution $A^i_{HR-MSI,L}$ is extracted

$$A^i_{HR-MSI,L} = \mathcal{P}\left(f^i_{HR-MSI,L}\right) \tag{7}$$

where $\mathcal{P}(\cdot)$ denotes a dual-stage convolutional mapping composed of two 3×3 kernels and ReLU activations. $A^i_{HR-MSI,L}$ captures the spatial distribution of local radiometric intensity. Due to its higher spatial resolution, the low-frequency component of HR-MSI preserves a more accurate spatial energy distribution and can therefore serve as a reliable spatial prior to calibrate the low-frequency radiometric energy distribution of LR-HSI.

Subsequently, a dynamic nonlinear refinement of $f_{LR-HSI,L}$ is implemented through spatial feature transform (SFT)

$$f^i_L = \gamma \odot f^i_{LR-HSI,L} + \beta \tag{8}$$

where γ and β are pixel-wise modulation maps learned from $A^i_{HR-MSI,L}$ via convolutional layers. This process performs pixel-wise nonlinear adjustment, compensating for spectral energy distribution deviations caused by insufficient spatial resolution, thereby producing high-fidelity spectral features f^i_L, as illustrated in Fig. 3.

2) Discriminative High-Frequency Feature Selection Module: DHFS leverages the structural benchmarks of LR-HSI to achieve noise suppression and the discriminative recomposition of HR-MSI high-frequency details. It takes the high-frequency pair $\{f^i_{HR-MSI,H}, f^i_{LR-HSI,H}\}$ as input. First, the structural benchmark $A^i_{LR-HSI,H}$ is extracted from the LR-HSI branch. Subsequently, a bank of deep convolutional operators $\{\mathcal{M}^i_j\}^n_{j=1}$ is introduced to perform differentiated gradient modulation on the HR-MSI high-frequency components, constructing multiple candidate feature streams C^i_j

$$C^i_j = \mathcal{M}_j\left(f^i_{HR-MSI,H}\right), \quad j \in \{1, 2, \ldots, n\} \tag{9}$$

where C^i_j explicitly expands the representation range of high-frequency details within the feature space. To identify high-fidelity details, $A^i_{LR-HSI,H}$ is employed as the consistency benchmark to evaluate the fitness score S^i_j for each candidate stream

$$S^i_j = \mathrm{CosSim}\left(C^i_j, A^i_{LR-HSI,H}\right) \cdot \exp\left(-\lambda \left\|\mathrm{Var}\left(C^i_j\right) - \mathrm{Var}\left(A^i_{LR-HSI,H}\right)\right\|_2\right) \tag{10}$$

where the cosine similarity (CosSim) constrains the directional consistency between each candidate stream and the LR-HSI high-frequency structure, while the variance (Var) penalty measures the discrepancy in high-frequency energy between the candidate stream and the LR-HSI benchmark. From a physical perspective, the variance difference reflects the deviation of the candidate stream from the reliable spatial energy distribution of the scene: a large discrepancy indicates that the high-frequency energy deviates from the true texture, whereas a small discrepancy preserves a texture distribution consistent with the LR-HSI. The exponential function ensures a smooth response for small deviations and stronger suppression for large deviations, thereby focusing on high-frequency features consistent with the scene structure. Finally, the candidate streams are aggregated via normalized weights $\omega^i_j = \mathrm{Softmax}(S^i_j)$ to obtain the enhanced and recomposed high-fidelity spatial features f^i_H

$$f^i_H = \sum_{j=1}^{n} \omega^i_j \cdot C^i_j. \tag{11}$$

As illustrated in Fig. 3, this process effectively suppresses high-frequency artifacts and noise through discriminative recomposition guided by physical structural priors, thereby obtaining more reliable spatial features.

D. FGSS Module

To effectively integrate the calibrated spatial and spectral information, we introduce the FGSS module, as illustrated in Fig. 2(c). Specifically, we concatenate f^i_L and f^i_H and map them into the full-frequency feature f^i_{LH}, which provides complementary semantic constraints to ensure the physical fidelity and structural consistency of the spatial–spectral information integration process. Subsequently, f^i_{LH} is utilized to guide the execution of a parallel cross-attention mechanism: global cross-attention (GCA) integrates spectral consistency features on a global scale, while local cross-attention (LCA) reinforces spatial details within local windows. GCA is defined as

$$z^i_G = \left[\mathcal{A}^i_G V^i_L, \mathcal{A}^i_G V^i_H\right] \tag{12}$$

where $[\cdot,\cdot]$ denotes the concatenation operation in the channel dimension, and $\mathcal{A}^i_G = \mathrm{Softmax}(Q^i_{LH}(K^i_{LH})^T / \sqrt{d_k})$ represents the computed global spatial–spectral semantic correlation weights. This mechanism suppresses the global spatial–spectral distortion caused by modality mismatch through long-range modeling.

Parallel to this, we introduce LCA. The LCA block first partitions the feature map into nonoverlapping local windows (size 8×8) via the window partitioning mechanism Win(·), and performs cross-attention computation within each window

$$z^i_L = \mathrm{WinRev}\left(\left[\mathcal{A}^{w,i}_L V^{w,i}_L, \mathcal{A}^{w,i}_L V^{w,i}_H\right]\right) \tag{13}$$

where WinRev(·) is the inverse operation of Win(·). Here, $\mathcal{A}^{w,i}_L$ follows the same calculation method as mentioned above, but the attention interaction is restricted within local windows. This branch leverages local consistency features to guide the interaction, enhancing edge sharpness while avoiding the texture smoothing caused by conventional local operators. Finally, the two branch outputs are fused through weighting with learnable scaling factors (η, ζ)

$$f^i_{fused} = \mathrm{Conv}_{1\times1}\left(\left[\eta \cdot z^i_G, \zeta \cdot z^i_L\right]\right). \tag{14}$$

FGSS is guided by full-frequency features to adaptively integrate global spectral information with local spatial details, thereby producing a more high-fidelity fused image.

E. Loss Function

$\mathbf{Y} \in \mathbb{R}^{H\times W\times C}$ and $\hat{\mathbf{Y}} \in \mathbb{R}^{H\times W\times C}$ denote the GT and reconstructed fused images, respectively. Under the influence of neural network "spectral bias" [55], in HMIF tasks, the model tends to preferentially fit low-frequency components

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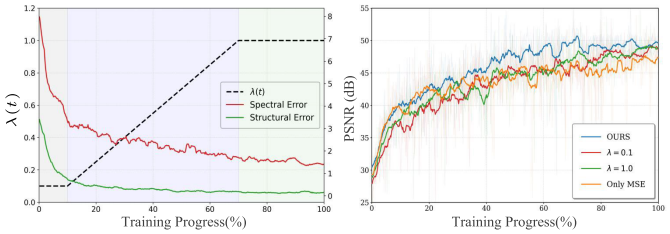


Fig. 4. Progressive supervision strategy analysis. (Left) Convergence curves of spatial and spectral errors under the regulation of $\lambda(t)$. In the early stage (gray region), the model prioritizes spectral basis stabilization; in the middle stage (purple region), $\lambda(t)$ increases linearly to guide spatial detail reconstruction; in the late stage (green region), the training enters a stable phase, achieving coordinated spatial–spectral optimization. (Right) PSNR training curves under different settings. Compared with the mse-only scheme and fixed-weight static strategies ($\lambda = 1.0$ and $\lambda = 0.1$), the proposed method (Ours) demonstrates faster convergence and higher reconstruction accuracy.

(mainly originating from the global spectra of LR-HSI) while underlearning high-frequency spatial details (primarily from HR-MSI). This bias often leads to spatial blurring and spectral distortion in the fused images. Using static losses (e.g., ERGAS or mse) is typically affected by this issue: during early training, the model prioritizes fitting the dominant low-frequency background to quickly reduce the overall loss, making it difficult to capture high-frequency details [22], [56]. To effectively balance spectral fidelity and spatial sharpness, we introduce a progressive supervision loss ($\mathcal{L}_{PS}$) in addition to the $\mathcal{L}_{mse}$ constraint. As illustrated in Fig. 4 (left), a dynamic annealing factor $\lambda(t)$ is employed to adaptively adjust the loss weighting. In the early training stage, the model primarily relies on the SAM loss to lock the spectral base; as training progresses, $\lambda(t)$ increases linearly, gradually emphasizing the SSIM loss to guide the reconstruction of spatial details on the stabilized spectral foundation. Compared to a single static loss, the progressive SAM+SSIM design effectively mitigates the "spectral bias," achieving an orderly integration from global spectral anchoring to local spatial refinement, thereby enhancing the overall spatio-spectral fidelity of the fused images. The specific computation is as follows:

$$\mathcal{L}_{PS} = \mathcal{L}_{SAM} + \lambda(t) \cdot \mathcal{L}_{SSIM} \tag{15}$$

where $\mathcal{L}_{SAM}$ is employed to constrain the spectral fidelity and is defined as

$$\mathcal{L}_{SAM}\left(\mathbf{Y}, \hat{\mathbf{Y}}\right) = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} \arccos\left(\frac{\mathbf{Y}_{i,j}^{\top} \hat{\mathbf{Y}}_{i,j}}{\left\|\mathbf{Y}_{i,j}\right\|_2 \left\|\hat{\mathbf{Y}}_{i,j}\right\|_2}\right) \tag{16}$$

where $\mathbf{Y}_{i,j}, \hat{\mathbf{Y}}_{i,j} \in \mathbb{R}^C$ denote the spectral vectors at pixel (i, j). The structural similarity loss ($\mathcal{L}_{SSIM}$) is designed to enhance high-frequency spatial structures and is defined as

$$\mathcal{L}_{SSIM}\left(\mathbf{Y}, \hat{\mathbf{Y}}\right) = 1 - \frac{1}{C} \sum_{c=1}^{C} SSIM\left(\mathbf{Y}_c, \hat{\mathbf{Y}}_c\right) \tag{17}$$

where $\mathbf{Y}_c, \hat{\mathbf{Y}}_c \in \mathbb{R}^{H \times W}$ represent the 2-D images corresponding to the cth spectral band. $\lambda(t)$ is dynamically adjusted over the course of training

$$\lambda(t) = \begin{cases} \lambda_{\min}, & t < t_s \\ \min\left(1, \lambda_{\min} + \dfrac{t - t_s}{t_e - t_s}\right), & t \geq t_s \end{cases} \tag{18}$$

where t_s and t_e are the start and end nodes of the annealing stage, respectively. In addition, the calculation formula for the mse loss is

$$\mathcal{L}_{mse} = \frac{1}{HWC} \left\|\hat{\mathbf{Y}} - \mathbf{Y}\right\|_2^2. \tag{19}$$

Finally, $\mathcal{L}_{PS}$ and $\mathcal{L}_{mse}$ are dynamically combined using learnable adaptive parameters α_i to construct the compound total loss function

$$\mathcal{L}_{\text{Total}} = \alpha_0 \cdot \mathcal{L}_{PS} + \alpha_1 \cdot \mathcal{L}_{mse}. \tag{20}$$

IV. Experiment

A. Datasets

To evaluate the effectiveness of PSFD-Net across various complex scenarios, four representative public hyperspectral remote sensing benchmark datasets are employed: Houston, Salinas, Pavia Center, and Pavia University. These datasets encompass diverse geographical features and spectral specifications, thereby ensuring the comprehensiveness and objectivity of the model evaluation in complex remote sensing environments.

1) Houston Dataset: The Houston dataset was acquired by the ITRES CASI-1500 hyperspectral sensor, capturing the urban landscape of the University of Houston campus and its surrounding areas in Texas. It contains 144 spectral bands ranging from 349 to 1046 nm, with a spatial dimension of 349 $\times$ 1905 pixels and a spatial resolution of 2.5 m. Due to its inclusion of various typical land-cover types such as urban structures, roads, and natural vegetation, this dataset is frequently utilized to assess the reconstruction accuracy of models in complex urban settings.

2) Salinas Dataset: The Salinas dataset was collected by the AVIRIS sensor over a continuous agricultural region in the Salinas Valley, California. The original data consist of 224 spectral bands, from which 204 effective bands are retained for experiments after removing those affected by noise and water vapor absorption. This dataset features a spatial resolution of 3.7 m and a dimension of 512 $\times$ 217 pixels. Owing to its rich variety of crop types and distinct spectral characteristics, it serves as a critical benchmark for verifying the spectral fidelity of models.

3) Pavia Center and Pavia University Dataset: The Pavia Center and Pavia University datasets were both acquired by the ROSIS sensor, corresponding to the densely populated city center and the university campus area in Pavia, Italy, respectively. Both datasets possess a high spatial resolution of 1.3 m. Pavia Center comprises 102 effective spectral bands with a dimension of 1096 $\times$ 715 pixels, while Pavia University retains 103 spectral bands with a dimension of 610 $\times$ 340 pixels. Benefiting from their high spatial precision and diverse land-cover distributions, these two datasets are widely adopted to validate the effectiveness of models in fine texture extraction and spatial reconstruction.

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TABLE I
QUANTITATIVE COMPARISON ON HOUSTON DATASET

Methods	RMSE↓	PSNR↑	ERGAS↓	SAM↓	SSIM↑	Params(M)	FLOPS(G)	Time(ms)
MSDCNN [41]	1.5644	44.0543	2.5461	2.8766	0.9841	1.649	27.017	4.213
SSFCNN [33]	1.2180	46.0258	3.3736	2.2205	0.9744	0.898	14.714	2.52
TFNet [42]	1.1594	46.4540	1.6503	1.9874	0.9848	2.485	9.69	0.921
SSRNET [34]	1.2059	46.1120	1.7256	2.2158	0.9813	0.56	9.172	3.925
MCT [45]	1.0592	47.2392	1.4434	1.8686	0.9867	6.935	25.181	13.167
MDC [35]	0.8175	49.4889	1.1854	1.4131	0.9903	34.64	47.098	17.794
AMSF [36]	1.2386	45.8799	1.6469	2.0859	0.9844	7.368	61.335	22.161
CYformer [47]	0.8080	49.5908	1.2231	1.4657	0.9913	0.87	14.971	7.31
Ours	0.7742	49.9613	1.1194	1.2788	0.9921	8.01	41.36	15.1

TABLE II
QUANTITATIVE COMPARISON ON SALINAS DATASET

Methods	RMSE↓	PSNR↑	ERGAS↓	SAM↓	SSIM↑	Params(M)	FLOPS(G)	Time(ms)
MSDCNN [41]	2.3791	39.9441	2.9526	1.7880	0.7994	2.229	36.528	5.38
SSFCNN [33]	1.8795	41.9913	8.8923	1.7214	0.8017	1.799	29.469	3.052
TFNet [42]	1.7978	42.3744	2.2862	1.3943	0.8372	2.537	10.54	0.956
SSRNET [34]	2.1483	40.8306	5.4396	1.8227	0.7713	1.124	18.409	1.147
MCT [45]	2.0372	41.2941	5.1777	1.5975	0.8161	12.636	49.553	13.168
MDC [35]	1.7029	42.8483	3.6782	1.4363	0.8286	69.295	93.908	18.816
AMSF [36]	2.3617	40.0076	5.5984	1.7134	0.7896	7.984	70.926	20.396
CYformer [47]	1.4162	44.4496	2.5594	1.2325	0.8531	1	17.349	7.06
Ours	1.3809	44.6693	2.5311	1.2156	0.8709	9.26	68.4	15.29

B. Experimental Setup

1) Data Preprocessing: In the data preprocessing phase, a rigorous sample partitioning and degradation strategy is implemented. For each dataset, a 128 × 128 pixel region is cropped from the central area to serve as the test set, while the remaining regions are utilized for training. During each training iteration, 128 × 128 pixel patches are randomly cropped from the training area to act as the GT. Subsequently, spatial degradation of the GT is performed using a 5×5 Gaussian blur kernel, and spatial downsampling is then applied using a scaling factor of 4 to generate LR-HSI. Simultaneously, spectral degradation of the GT is performed, and a uniform interval sampling strategy is used to extract five representative bands from the real image to generate HR-MSI, which serve as the inputs for model training and validation.

2) Experimental Environment Settings: PSFD-Net and all comparison methods were implemented on a computing platform equipped with an NVIDIA GeForce RTX 4090 GPU, utilizing the PyTorch 2.0.0 framework and CUDA 12.7 environment. Each training iteration randomly selects a 128×128 pixel block from the training area for training, with a total of 10 000 training iterations. Model optimization uses the Adam optimizer. We initialize the learnable scaling factor in FGSS to $\eta = \zeta = 0.5$, the learnable parameters of the loss function to $\alpha_0 = \alpha_1 = 0.5$, and the initial learning rate to 1×10^{-4}. To ensure the fairness of the comparative experiments, all methods were executed under identical hardware conditions and trained strictly according to their default configurations.

3) Evaluation Metrics: To objectively evaluate the performance of the models in terms of spatial structure restoration and spectral fidelity, five mainstream quantitative metrics are selected. Specifically, these include: root mean square error (RMSE) and peak signal-to-noise ratio (PSNR), which are used to measure the pixel-wise error and signal-to-noise ratio levels between the reconstructed and reference images; structural similarity index (SSIM), which assesses structural preservation across the three dimensions of luminance, contrast, and texture; erreur relative globale adimensionnelle de synthése (ERGAS), employed to measure the overall quality of the reconstruction results on a global scale; and spectral angle mapper (SAM), utilized to characterize the angular difference between spectral vectors, serving as a key metric for evaluating

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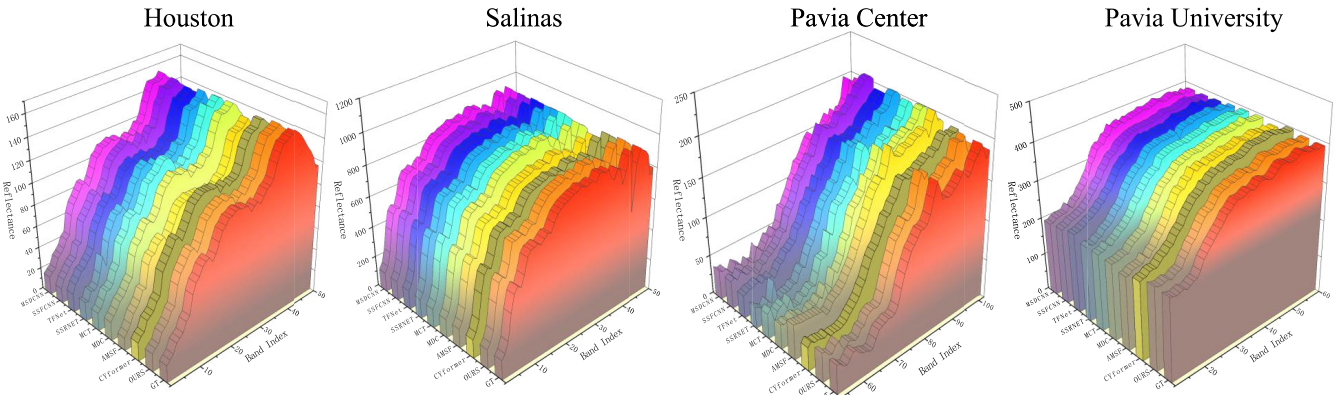


Fig. 5. Spectral reflectance curves of 50 randomly selected bands at the spatial location (80, 80) across four datasets for different models. Compared with other methods, OURS produces spectral curves that align most closely with the GT, demonstrating superior spectral fidelity.

TABLE III
QUANTITATIVE COMPARISON ON PAVIA DATASET

Methods	RMSE↓	PSNR↑	ERGAS↓	SAM↓	SSIM↑	Params(M)	FLOPS(G)	Time(ms)
MSDCNN [41]	3.4609	37.3469	3.9300	4.2764	0.9832	1.242	20.358	3.802
SSFCNN [33]	3.2149	37.9841	3.7132	3.7411	0.9854	0.452	7.404	1.437
TFNet [42]	3.2627	37.8594	3.7112	4.2800	0.9817	2.448	9.095	0.864
SSRNET [34]	3.2661	37.8503	3.7254	3.7257	0.9850	0.281	4.602	0.634
MCT [45]	2.8099	39.1571	3.1503	3.7569	0.9860	4.017	13.067	8.065
MDC [35]	2.9177	38.8478	3.3424	3.7399	0.9859	17.461	23.876	14.156
AMSF [36]	3.4098	37.4761	3.6987	5.0685	0.9713	7.053	56.517	19.218
CYformer [47]	2.6994	39.5056	3.0357	3.5937	0.9863	0.78	13.306	7.52
Ours	2.6378	39.706	3.0049	3.6353	0.9864	7.83	21.01	12.45

spectral consistency. Through these complementary metrics, a comprehensive evaluation of PSFD-Net's performance can be conducted from multiple dimensions.

C. Comparison With State-of-the-Art Methods

1) Quantitative Comparisons: As shown in Table I, on the Houston dataset, PSFD-Net achieves an improvement of approximately 0.37 dB in PSNR over the second-best method CYformer, while attaining the lowest SAM value, indicating its ability to correct low-frequency energy shifts caused by insufficient spatial resolution. On the Salinas dataset (see Table II), despite the reconstruction challenges posed by contiguous crop fields, PSFD-Net maintains superior performance with an SSIM of 0.8709, demonstrating high structural fidelity. This suggests that the model, through constraints from global anchoring to local refinement, effectively balances spectral accuracy and spatial detail reconstruction. The spectral curve comparison in Fig. 5 further confirms that, on the Houston dataset, PSFD-Net closely matches the GT, validating the effectiveness of the frequency-domain decoupling and physical calibration strategies in disentangling and correcting complementary modal defects, thereby achieving high-fidelity reconstruction.

As reported in Table III, on the Pavia Center dataset, PSFD-Net attains the best performance in both RMSE and PSNR, while preserving high structural fidelity and achieving high-fidelity extraction and reconstruction of local details. On the Pavia University dataset (see Table IV), the performance advantage of PSFD-Net is particularly pronounced, with a PSNR of 44.0884 dB and an SSIM of 0.9829, while also achieving the lowest SAM value. This indicates that the model can effectively suppress noise contamination caused by spatial and spectral mixing when handling spectrally similar campus objects. The spectral curve of specific pixels in Fig. 5 further demonstrates the model's robustness in reconstructing edge and heterogeneous objects.

2) Qualitative Comparisons: On the Houston dataset (see Fig. 6), the comparison methods generally exhibit noticeable artifacts or spatial blurring along dense building edges. In contrast, OURS produces building boundaries that are sharp and geometrically consistent, clearly outperforming other approaches. The enlarged difference maps further show that OURS yields the most uniform and smooth residuals, depicted in deep blue, demonstrating its superior reconstruction accuracy in preserving spatial details. On the Salinas dataset (see Fig. 7), OURS achieves the best visual quality,

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TABLE IV
QUANTITATIVE COMPARISON ON PAVIAU DATASET

Methods	RMSE↓	PSNR↑	ERGAS↓	SAM↓	SSIM↑	Params(M)	FLOPS(G)	Time(ms)
MSDCNN [41]	2.2135	41.3345	1.6068	2.528	0.9519	1.252	20.517	3.806
SSFCNN [33]	2.4321	40.7012	2.3431	2.816	0.9412	0.461	7.55	1.469
TFNet [42]	1.8218	42.6057	1.3528	2.1426	0.9757	2.449	9.109	0.871
SSRNET [34]	1.6278	43.5835	1.2090	1.9473	0.9807	0.286	4.6939	0.642
MCT [45]	1.5696	43.9001	1.1736	1.8874	0.9815	4.074	13.276	8.238
MDC [35]	1.5581	43.9639	1.1737	1.8665	0.9816	17.785	24.286	15.278
AMSF [36]	1.9789	41.6901	1.4806	2.4026	0.9747	7.059	56.614	19.221
CYformer [47]	1.5605	43.9507	1.1729	1.8700	0.9821	0.782	13.346	10.82
Ours	1.5359	44.0884	1.1561	1.8521	0.9829	7.97	21.36	12.31

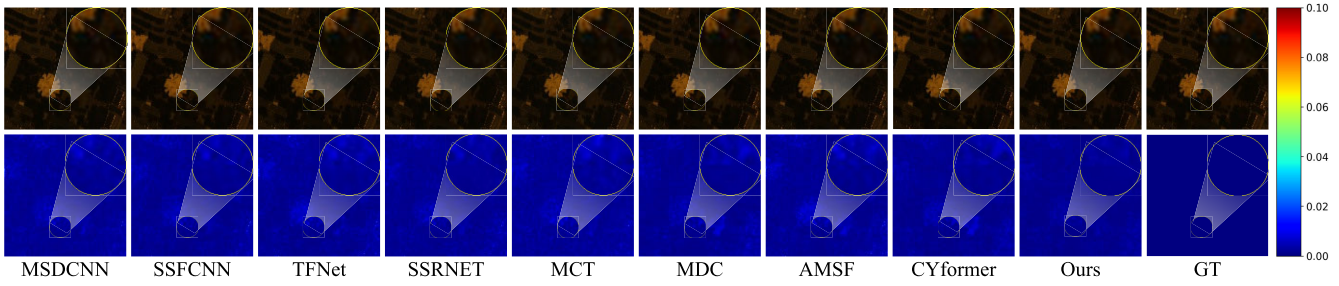


Fig. 6. Fusion results of different methods on the Houston dataset. The first row shows the fused pseudo-color RGB images, with a selected local region magnified and indicated by a circle. The second row shows the pseudo-color difference images, with a magnified view of a selected local region of the difference map indicated by a circle.

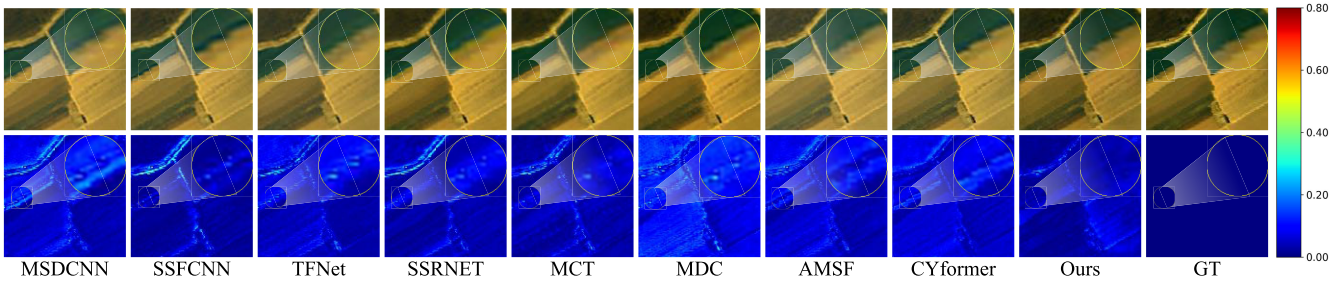


Fig. 7. Fusion results of different methods on the Salinas dataset. The first row shows the fused pseudo-color RGB images, with a selected local region magnified and indicated by a circle. The second row shows the pseudo-color difference images, with a magnified view of a selected local region of the difference map indicated by a circle.

successfully restoring the jagged edges of farmland and realizing high-fidelity fusion, highlighting its precise reconstruction capability along object boundaries.

On the Pavia Center dataset (see Fig. 8), comparison methods often exhibit ringing artifacts or loss of fine details along building shadow edges. Leveraging the frequency-domain decoupling mechanism, OURS precisely separates and enhances high-frequency textures, enabling the reconstructed images to maintain clear boundaries even in dense urban areas. Examination of local zoom-ins and corresponding difference maps reveals nearly uniform deep-blue residuals, visually confirming the model's outstanding ability to restore complex spatial structures. On the Pavia University dataset (see Fig. 9), which includes grass, asphalt roads, and spectrally similar rooftops, OURS demonstrates high sharpness at object boundaries. Local magnifications show clearly delineated edges, and the difference maps exhibit minimal structural errors, further validating the effectiveness of OURS in complex scenes.

3) Computational Efficiency Comparisons: To further assess the practical feasibility of PSFD-Net in large-scale remote sensing applications, we evaluated its computational efficiency on four datasets (see Tables I–IV). The results demonstrate that PSFD-Net achieves a favorable balance between high-fidelity reconstruction and computational cost.

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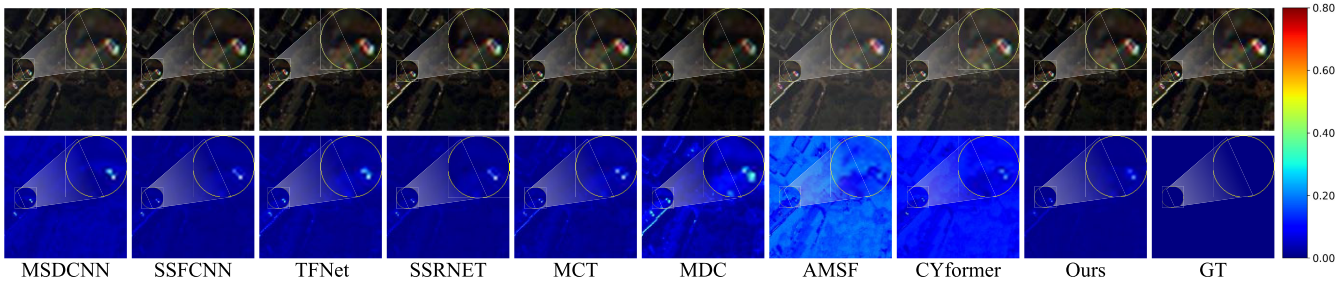


Fig. 8. Fusion results of different methods on the Pavia Center dataset. The first row shows the fused pseudo-color RGB images, with a selected local region magnified and indicated by a circle. The second row shows the pseudo-color difference images, with a magnified view of a selected local region of the difference map indicated by a circle.

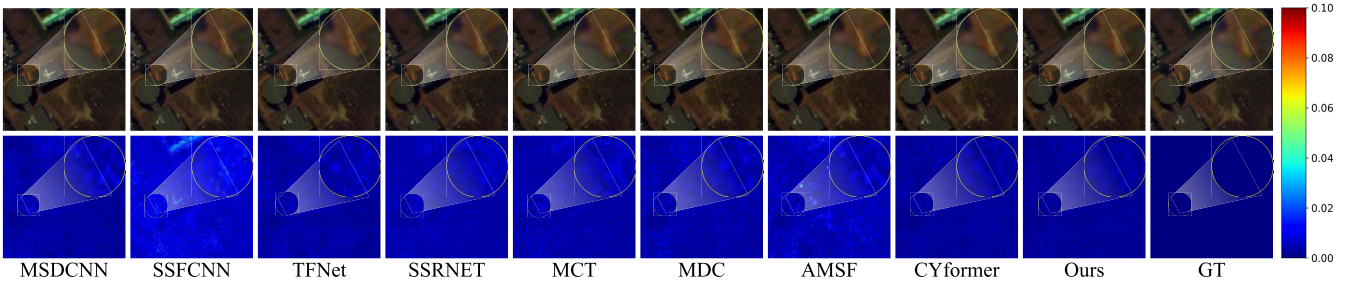


Fig. 9. Fusion results of different methods on the Pavia University dataset. The first row shows the fused pseudo-color RGB images, with a selected local region magnified and indicated by a circle. The second row shows the pseudo-color difference images, with a magnified view of a selected local region of the difference map indicated by a circle.

TABLE V
MODULE ABLATION EXPERIMENT RESULTS OF PSFD-NET ON HOUSTON DATASET

AFDM	PAFC	FGSS	RMSE	PSNR	ERGAS	SAM	SSIM
		✓	1.0524	48.0923	1.3353	1.7090	0.9884
	✓	✓	1.2124	48.5422	1.3212	1.8686	0.9881
✓		✓	1.0128	48.6079	1.6032	1.4654	0.9870
✓	✓		1.0122	48.6996	1.3012	1.6079	0.9895
✓	✓	✓	**0.7742**	**49.9613**	**1.1194**	**1.2788**	**0.9921**

Compared with high-performance state-of-the-art models such as MDC, PSFD-Net maintains a streamlined architecture while delivering superior fusion accuracy. For instance, on the Salinas dataset (see Table II), its number of parameters and FLOPs are significantly reduced, effectively alleviating the hardware burden associated with large multiscale networks. Although some lightweight CNN-based models incur lower computational overhead, their reconstruction performance in terms of spectral and spatial consistency is markedly inferior to that of PSFD-Net. Moreover, PSFD-Net achieves shorter inference time, consistently outperforming advanced methods such as AMSF. This excellent performance–cost ratio highlights the scalability and robustness of PSFD-Net for efficient and high-accuracy processing in real-world remote sensing scenarios.

D. Ablation Study

To verify the effectiveness of each component in PSFD-Net, we conducted ablation studies on the Houston and Pavia Center datasets to comprehensively assess the contribution of each module to the overall model performance.

We conducted module ablation experiments on the Houston dataset to evaluate the contributions of the key components in PSFD-Net, including AFDM, PAFC, and FGSS, as summarized in Table V. First, removing the frequency-domain decoupling and calibration mechanisms (AFDM and PAFC) and directly concatenating the original multiscale features, followed by convolutional dimensionality reduction, led to a significant drop in performance, indicating that frequency-domain decoupling and calibration effectively mitigate spatial and spectral distortions and substantially enhance fusion accuracy. Subsequently, replacing AFDM with spatial convolution and average pooling for feature extraction also resulted in performance degradation, particularly in ERGAS and SSIM, demonstrating that frequency-domain processing more accurately captures global spectral information and local details. Replacing PAFC with simple intramodal feature concatenation caused a decrease in PSNR due to the absence of a physically aware cross-modal calibration mechanism, validating the effectiveness of PAFC in low-frequency radiation alignment and high-frequency discriminative selection. Finally, substituting FGSS with simple feature addition disrupted the balance between detail reconstruction and global spectral enhancement, leading to increased RMSE and decreased PSNR, highlighting the critical role of FGSS's full-frequency guided global-LCA in integrating spatial–spectral features.

1) Effectiveness of PAFC: To further evaluate the contributions of the individual submodules within PAFC, we conducted ablation experiments on its internal structure, with results summarized in Table VI. Completely removing PAFC prevents the decoupled features from undergoing cross-modal calibration, significantly degrading fusion performance, indicating

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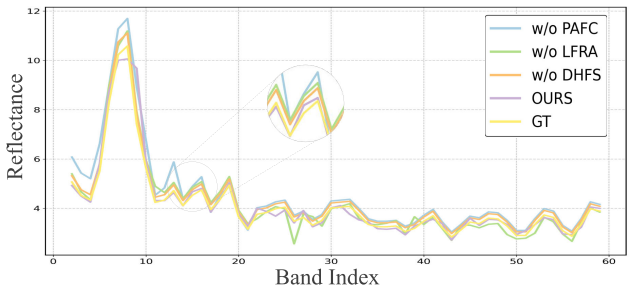


Fig. 10. Comparison of spectral reflectance curves at coordinate (50, 50) on the Pavia Center dataset. When PAFC, LFRA, or DHFS is removed, noticeable spectral deviations occur. In contrast, the curves produced by OURS closely match GT.

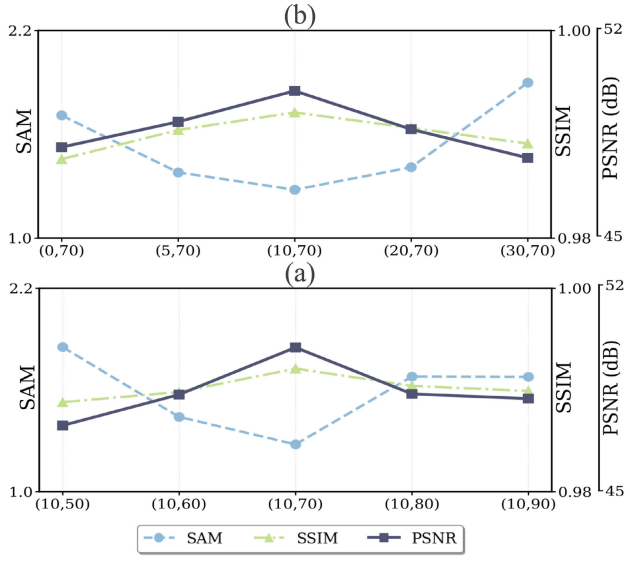


Fig. 11. Analysis of the progressive supervision parameters on the Houston dataset. The plots illustrate the impact of (a) supervision endpoint t_e (with fixed t_s = 10%) and (b) supervision start-point t_s (with fixed t_e = 70%) on PSNR, SSIM, and SAM. The results indicate that premature structural forcing or insufficient transition time leads to a performance drop, with (t_s, t_e) = (10%, 70%) achieving the optimal spectral–spatial balance.

TABLE VI
ABLATION EXPERIMENT RESULTS OF PAFC ON HOUSTON DATASET

	RMSE	PSNR	ERGAS	SAM	SSIM
w/o PAFC	1.0128	48.6079	1.6032	1.4654	0.9870
w/o LFRA	0.8187	49.5089	1.3453	1.6089	0.9910
w/o DHFS	0.8109	49.6181	1.4021	1.3765	0.9902
w/ PAFC	**0.7742**	**49.9613**	**1.1194**	**1.2788**	**0.9921**

TABLE VII
ABLATION EXPERIMENT RESULTS OF FGSS ON HOUSTON DATASET

	RMSE	PSNR	ERGAS	SAM	SSIM
w/o f_{LH}^i	0.8988	49.1689	1.5423	1.5312	0.9909
w/ f_{LH}^i	**0.7742**	**49.9613**	**1.1194**	**1.2788**	**0.9921**

that PAFC plays a crucial role in achieving physically aware alignment across modalities. Replacing the LFRA module with a standard convolution block results in insufficient calibration of low-frequency features and a notable increase in SAM, confirming the importance of LFRA in maintaining global spectral consistency and mitigating spectral shift. Similarly, substituting the Discriminative DHFS module with a convolution block leads to a clear deterioration in ERGAS and SSIM, demonstrating that DHFS is essential for high-frequency feature selection and detail reconstruction. As shown in Fig. 10, removing PAFC, LFRA, or DHFS introduces varying degrees of spectral deviation, whereas the full model (OURS) produces a spectral curve that closely matches the GT, further validating the critical role of PAFC and its submodules in full-spectrum spectral calibration and enhancement.

2) Effectiveness of $\mathcal{L}_{PS}$: To validate the effectiveness of the progressive supervision strategy, we compare the proposed $\mathcal{L}_{PS}$ with the mse-only loss (i.e., without introducing $\mathcal{L}_{PS}$) and fixed-weight settings (λ = 0.1 and 1.0). As shown in Fig. 4 (right), although the mse-only scheme achieves faster PSNR improvement in the early training stage, the lack of physical constraints limits its capability in modeling fine structural details, leading to noticeable oscillations in the later stage and the lowest final convergence accuracy. The fixed-weight strategies exhibit slower convergence and limited reconstruction accuracy. Specifically, λ = 1.0 introduces excessively strong structural constraints at the early stage, which interferes with the stable establishment of the spectral basis, whereas λ = 0.1 provides insufficient structural guidance and fails to support high-precision fusion reconstruction. In contrast, Ours achieves a smooth transition from spectral anchoring to spatial refinement through adaptive weight adjustment, resulting in more stable training behavior in the later stage and superior convergence accuracy. These results demonstrate that the proposed strategy provides a better balance between spectral fidelity and spatial detail reconstruction.

3) Effectiveness of Parameter in $\mathcal{L}_{PS}$: To validate the reasonableness of the start (t_s) and end (t_e) points of the annealing mechanism in $\mathcal{L}_{PS}$, we conducted an ablation study, as shown in Fig. 11. When t_s is fixed, varying t_e from 50% to 90% results in a rise-then-fall trend in performance: a too-low t_e imposes structural constraints prematurely while the spectral mapping is not yet stable, overemphasizing local structures and hindering spectral optimization, which prevents PSNR from reaching its optimum; a too-high t_e prevents the model from fully correcting complex spatial details within limited iterations, leading to decreased SSIM and PSNR. When t_e is fixed, increasing t_s from 0% to 30% shows a similar rise-then-fall trend: a too-low t_s triggers structural remodeling before the spectral features are stable, interfering with spectral learning and limiting SAM improvement; a too-high t_s results in a rapid change of loss weights, forcing the model to switch from spectral learning to structural remodeling in a short time, thereby preventing the spectral–spatial balance from reaching its optimum.

4) Effectiveness of the FGSS: To further verify the effectiveness of the full-frequency guidance mechanism within the FGSS module, we constructed a baseline network for comparison by removing the guidance branch while retaining only the parallel cross-attention backbone. The ablation results, as

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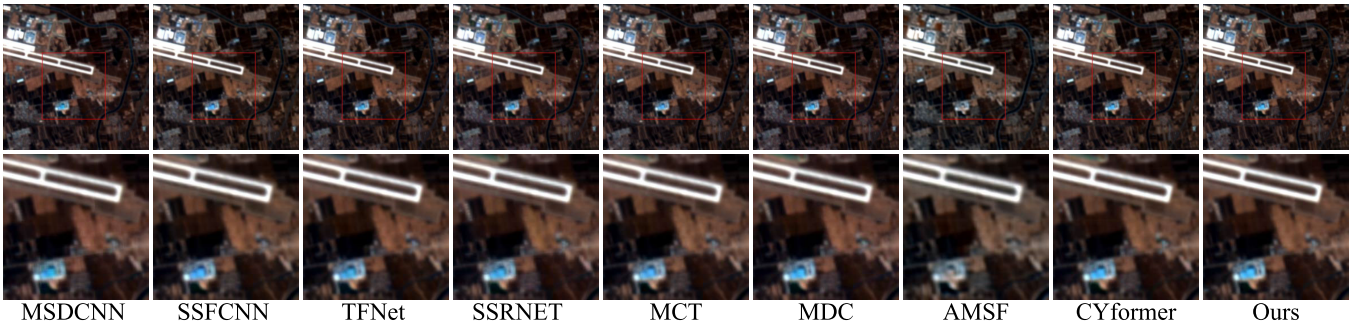


Fig. 12. Fusion results of different methods on the GF5-GF6 dataset. The first row displays the fused pseudo-color RGB images, and the second row provides the magnified views of the subregions demarcated by red boxes for a clearer comparison of spatial details.

TABLE VIII
QUANTITATIVE NO-REFERENCE EVALUATION OF GF5–GF6 HSI–MSI FUSION RESULTS

Metrics	MSDCNN [41]	SSFCNN [33]	TFNet [42]	SSRNET [34]	MCT [45]	MDC [35]	AMSF [36]	CYformer [47]	Ours
BRISQUE [58]	74.7186	78.2977	75.6562	76.3914	77.4328	76.3985	76.2286	74.9637	**74.5567**
NSSQS [59]	33.9853	35.6414	36.4576	36.2482	35.1668	34.25432	35.2109	34.4237	**32.3913**

reported in Table VII, demonstrate that the inclusion of full-frequency guidance leads to a significant improvement in key metrics such as SAM and SSIM compared to the baseline model. This provides strong evidence that the full-frequency features f_{LH}^i, generated by fusing f_L^i and f_H^i, provide crucial cross-modal semantic constraints during feature integration. Through this guidance mechanism, GCA and LCA can achieve precise adaptive alignment between global spectral consistency and local spatial details, effectively suppressing nonphysical artifacts caused by feature mismatch. Consequently, this ensures a deep synergy between spatial structural reconstruction and spectral fidelity.

5) Experiment on Real-Life HSI–MSI Fusion: To further validate the effectiveness of the proposed method in practical applications, we conduct performance evaluations on a real-world satellite dataset. The experimental data are acquired by the GaoFen-5 (GF-5) and GaoFen-6 (GF-6) satellites. Specifically, the LR-HSI is captured by the Visible and Short-Wave Infrared Hyperspectral Imager (AHSI) on the GF-5 satellite, comprising 330 spectral bands with a wavelength range of 400–2500 nm. The HR-MSI is obtained by the wide field of view (WFV) sensor on the GF-6 satellite, featuring four effective bands across the 450–900 nm spectral range. To ensure experimental fairness and reproducibility, all preprocessing procedures—including registration, cropping, and normalization—strictly follow the settings in [59] without any additional processing.

Fig. 12 illustrates the fusion results for the GF-5/GF-6 data pair. As observed from the magnified regions highlighted by red boxes, PSFD-Net recovers sharper structural edges and more natural texture details compared to other state-of-the-art methods. For instance, in local building areas, several competing methods exhibit noticeable color shifts, indicating spectral distortion. In contrast, the proposed method effectively preserves structural details while maintaining superior spectral consistency, resulting in high-fidelity fused images. Since real-world satellite data lack HR-HSI for GT, full-reference metrics cannot be employed for quantitative evaluation. Consequently, we utilize no-reference quality assessment metrics to evaluate the fusion results, specifically the blind/referenceless image spatial quality evaluator (BRISQUE) [57], and the no-reference spectral–spatial quality score (NSSQS) [58]. Note that lower values indicate better performance for both BRISQUE (spatial naturalness and lower distortion) and NSSQS (better spectral–spatial consistency). As shown in Table VIII, PSFD-Net achieves the best results across both no-reference metrics, demonstrating its capability to enhance spatial quality and spectral consistency in real-world scenarios. This further validates the superiority of the proposed method for practical applications.

V. CONCLUSION

To accurately extract complementary features and achieve high-fidelity reconstruction, this study proposes PSFD-Net. The model employs AFDM to decompose features in the frequency domain, effectively capturing spatial and spectral information and mitigating the coupling of features in the spatial domain. Experiments indicate that the decoupled spatial–spectral features still exhibit energy bias; to address this, PAFC is introduced for spatial–spectral calibration, effectively mitigating distortions caused by sensor spatial degradation and spectral undersampling. Subsequently, FGSS is used to integrate the spatial–spectral features, enabling consistent reconstruction of spatial structures and spectral information under progressive supervision. Extensive experiments on four public datasets demonstrate that PSFD-Net significantly outperforms existing methods in both quantitative metrics and visual fidelity.

However, despite the superior performance of our method, certain limitations remain. Specifically, for hyperspectral data with a large number of bands, the global-LCA architecture still faces computational bottlenecks. As the number of spectral bands increases, the complexity of the self-attention mechanism grows quadratically, posing challenges

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for resource-constrained environments. Future work will investigate the use of Linear Attention or the Mamba architecture to substantially improve computational efficiency while maintaining high-fidelity reconstruction.

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